

TUNIR SIKDAR

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer, with almost 6 years of experience architecting and scaling distributed data platforms on AWS and Azure. Skilled at translating ambiguous data problems, missing feeds, inconsistent vendor formats, historical gaps, into fault-tolerant, idempotent pipeline designs and Lakehouse architectures supporting high-volume, mission-critical workloads. Owns system design end-to-end, improving reliability, performance, and cost efficiency for Fortune 500 clients across insurance, retail, and e-commerce.

SKILLS

- **Languages:** Python, SQL, T-SQL
- **System Design:** Distributed systems, Lambda & Kappa architecture, Medallion architecture, SLA-driven orchestration, scalability & process optimization
- **AWS:** Lambda, Glue, S3, EC2, CloudWatch, Secrets Manager, SNS
- **Azure:** Databricks, Data Factory (ADF), Data Lake Storage (ADLS Gen2), Key Vault, Azure SQL Database, Event Hubs
- **Big Data:** Apache Spark, PySpark, Spark SQL, Apache Iceberg, Hadoop (basic)
- **Data Engineering:** ETL/ELT design, data modeling & normalization, entity relationship diagrams, Data Warehousing, Data Vault 2.0, data governance, data lineage, data quality frameworks
- **DevOps & Tools:** Azure DevOps, Git, CI/CD
- **MLOps:** Basic exposure to model deployment pipelines and monitoring

EXPERIENCE

Data Engineer II | Deloitte USI - Kolkata, India

04/2025 - Present

- Architected a greenfield, config-driven ingestion framework with idempotent backfill logic, scaling to 4.2 TB of daily volume with zero data loss.
- Built technical and functional data quality checks with DLQ routing and imputation rules to fix malformed vendor fields.
- Led a team of 5 engineers to overhaul commission payout processing from weekly to daily, cutting manual effort by ~90% and chargeback turnaround from a month to a day.
- Architected a fault-tolerant orchestration layer on AWS Lambda with automatic retry, SNS alerting, and full audit trails sustaining a 99%+ on-time SLA.
- Implemented PII data masking and column-level access controls using Databricks Unity Catalog for centralized governance, enabling secure RBAC-based data access across teams.

Firm Initiatives

- Built an event ingestion pipeline for a mobile app using Azure Event Hubs and Spark Structured Streaming, tracking user health metrics and usage history for stakeholders.

Engineer | Fractal Analytics - Pune, India

09/2023 - 03/2025

- Delivered 24 end-to-end Big Data projects on Azure, designing scalable PySpark and Spark SQL jobs on datasets exceeding 500 GB and improving efficiency by 40%.
- Fixed recurring data gaps in a Fortune 500 retail client's inventory feeds with config-driven ETL pipelines, designing and implementing a snowflake schema data warehousing solution using Delta tables for reliable BI reporting.
- Designed automated data validation and monitoring frameworks in PySpark, cutting manual error handling by 60%, and re-architected workflows onto Azure to save \$150K annually.

Application Developer | IBM India Pvt. Ltd. - Kolkata, India

02/2021 - 08/2023

- Migrated 40+ business-critical pipelines from legacy systems to Azure using ADF.
- Built data quality validation checks in Databricks and Python.
- Streamlined data extraction processes, saving 42 hours per month and cutting operating costs by 30%.

EDUCATION

B.Tech. in Electronics and Communication Engineering, Heritage Institute of Technology

08/2016 - 08/2020

GPA: 9.09/10.0

CERTIFICATIONS

- Databricks Certified Data Engineer Associate (2025)
- Fractal Certified PySpark Developer

AWARDS

- Outstanding Award & Applause Award, Deloitte USI | Star Award (Kaizen), Fractal Analytics | Manager Appreciation Award, IBM India Pvt. Ltd.